

INFINITY FITNESS STUDIOS

Power BI Requirements

Member Retention, Utilization & Operational Analytics
December 2025 - February 2026

Document Control	Details
Document ID	IFS-PBI-001
Version	1.0
Role	Data Analyst
Primary User	Gym Manager
Purpose	Define the reporting, interaction, data and validation requirements for the Power BI solution built for the Infinity Fitness analytics engagement.
Source Basis	IFS-BRD-001, IFS-USAC-001, IFS-KPI-001, IFS-DD-001 and IFS-ASM-001.
Confirmed Build Fact	The Power BI dashboard was built from scratch.
Historical Design Control	Exact original page count, page names, visual layout, colors and DAX implementation are not retained. This document specifies defensible functional requirements without pretending to reproduce the original dashboard exactly.

1. Reporting Objective

The Power BI solution was intended to convert prepared membership and operational data into a management-facing analytical view. It needed to make retention, attrition, cohort behavior, attendance/utilization, renewals, plan performance, trainer activity and relevant commercial information easier to investigate than the original unclear Excel records.

2. Intended Users & Decisions

User	Information Need	Decision Supported
Gym Manager	Retention, attrition and engagement patterns	Retention and re-engagement actions
Gym Manager	Renewal and plan behavior	Pricing, renewal incentives and plan discussions
Gym Manager	Attendance by hour/day	Peak-period and underutilized-period operating decisions
Gym Manager	Trainer sessions, bookings, workload, assignments and attendance	Trainer/staff allocation discussions
Gym Manager	Revenue and expense visibility where supported	Commercial performance review

3. Functional Reporting Requirements

ID	Requirement	Priority*
PBI-FR-01	Provide a consolidated management view of the key analytical areas supported by the prepared dataset.	High
PBI-FR-02	Display member retention and attrition/churn according to the project definitions and selected analysis context.	High
PBI-FR-03	Support joining-month cohort analysis across subsequent attendance/retention periods.	High
PBI-FR-04	Show attendance and membership-usage patterns over time.	High
PBI-FR-05	Enable identification of peak and underutilized periods using attendance/check-in data.	High
PBI-FR-06	Provide renewal behavior and membership-plan comparison where the underlying data supports it.	High
PBI-FR-07	Provide trainer-utilization views using sessions, bookings, workload, assignments and attendance where available.	Medium-High
PBI-FR-08	Provide revenue and expense visibility where validated source data supports the analysis.	Medium
PBI-FR-09	Allow relevant filters to change the analytical context consistently across applicable visuals.	High
PBI-FR-10	Ensure dashboard results can be reconciled to the prepared analytical data.	High
PBI-FR-11	Exclude confidential member-level data and unsupported client values from public portfolio versions.	High

*Priority is a portfolio BA prioritization derived from confirmed business objectives, not a claimed historical client ranking.

4. Recommended Logical Dashboard Views

Because the exact historical page names and count are not retained, the following are logical portfolio views, not claims about the original dashboard structure.

Logical View	Purpose	Primary Content
Executive Overview	Provide a concise management summary.	Member/renewal indicators, attendance/utilization signals and relevant commercial context.
Retention & Attrition	Investigate member continuation and disengagement.	Retention, attrition, renewal, active-member and cohort views.
Utilization Analysis	Understand facility usage patterns.	Check-ins, attendance frequency, time-of-day/day patterns, peak and underutilized periods.
Plan & Renewal Analysis	Compare membership behavior.	Plan membership, renewal behavior and plan-level comparisons.
Trainer Utilization	Understand trainer activity.	Sessions, bookings, workload, assignments and attendance.
Commercial Analysis	Review validated financial information.	Revenue and expense trends where source data supports them.

5. KPI Requirements

KPI Group	Metrics / Analytical Indicators
Member	Member Retention Rate; Member Attrition/Churn Rate; Membership Renewal Rate; Active Member Count; Cohort Retention
Utilization	Total Check-ins/Attendance; Attendance Frequency per Member; Peak Period Utilization; Underutilized Periods
Plan / Commercial	Plan Membership Count; Plan Renewal Rate; Revenue; Expenses
Trainer	Trainer Session Count; Trainer Booking/Attendance Volume; Trainer Workload

Exact metric formulas must follow the KPI Catalogue and verified source logic. Where the original calculation detail is not retained, the dashboard must not present a reconstructed formula as historical fact.

6. Filtering & Interaction Requirements

- Provide period/date filtering where the prepared data supports consistent time analysis.
- Allow membership-plan filtering for plan and renewal comparisons.
- Allow joining-cohort or equivalent cohort context for retention analysis.
- Allow trainer filtering for trainer-utilization analysis where trainer identifiers are available.
- Ensure filters affect only relevant visuals and do not create misleading cross-filter behavior.
- Use clear visual titles and context so the manager can understand what a filtered result represents.

7. Visual Requirements

Analytical Need	Suitable Visual Treatment	Requirement
Headline performance	Cards / KPI indicators	Use for concise measures that have a verified calculation and meaningful selected-period context.
Trend over time	Line or column trend	Show attendance, renewal, revenue/expense or other time-based movement where supported.
Cohort retention	Matrix / heatmap-style cohort view	Show joining cohorts against subsequent periods without exposing confidential public values.
Plan comparison	Bar/column or comparison table	Compare plan membership/renewal behavior without implying causation.
Peak-hour analysis	Time-based column/line view	Make higher- and lower-use periods distinguishable; evening activity should be visible where supported by the data.
Trainer activity	Bar/table/matrix	Compare sessions, bookings, assignments, attendance or workload indicators.
Detailed review	Table/matrix	Provide summarized analytical detail without exposing restricted member-level information in public outputs.

8. Data & Model Requirements

- Use prepared data representing the confirmed member, membership, attendance, payment, trainer, class/session, revenue and expense domains as applicable.
- Use stable logical identifiers to link analytical entities and prevent unintended many-to-many duplication where possible.
- Create a consistent date/time analysis structure for joining cohorts, renewals, attendance and period trends.
- Standardize membership-plan categories before plan-level reporting.
- Validate relationships so member, attendance, payment and trainer/session joins do not inflate metrics.
- Keep derived fields such as joining month, membership duration, renewal flag, attrition flag and time band traceable to source logic.

9. Data Quality Requirements

Area	Power BI Requirement
Duplicates	Duplicate source records that could inflate measures must be resolved or excluded before reporting.
Missing Data	Missing renewal/cancellation/activity evidence must not automatically be classified as churn.
Dates / Times	Invalid or inconsistent date/time values must be addressed before cohort or utilization reporting.
Plan Labels	Equivalent plan labels must be standardized before comparison.
Amounts	Revenue, payment and expense fields must be validated as numeric and reconciled before aggregation.
Relationships	Unmatched or duplicate key relationships must be reviewed before measures are relied upon.

10. Dashboard Validation Requirements

Validation ID	Check	Expected Result
VAL-01	Compare headline totals with prepared analytical data.	Dashboard totals reconcile at the agreed analysis grain.
VAL-02	Apply date/period filters.	Applicable visuals update consistently for the selected period.
VAL-03	Apply plan filters.	Plan-related metrics and visuals reflect the selected plan context.
VAL-04	Review joining cohorts.	Members are assigned to the correct joining-month cohort and subsequent periods.
VAL-05	Review attendance by time.	Peak/underutilized views reconcile to valid check-in date/time data.
VAL-06	Review trainer views.	Trainer metrics reconcile to available session/booking/assignment/attendance records.
VAL-07	Review financial views.	Displayed revenue/expense values reconcile to validated analytical totals.
VAL-08	Review public portfolio version.	No restricted member data or unsupported confidential values are exposed.

11. Usability Requirements

- Keep the management view readable without requiring the user to inspect raw Excel sheets.
- Use consistent terminology across retention, attrition, renewal, utilization and plan analysis.
- Use concise titles and labels that explain the business question represented by each visual.
- Avoid overcrowding a page with redundant charts or decorative visuals that do not support a decision.
- Make filter state understandable so the manager can interpret the displayed context correctly.

12. Confidentiality & Portfolio Requirements

- Do not publish raw client/member data.
- Do not publish commercially sensitive revenue, expense, retention or churn values when restricted or no longer retained.
- Do not create synthetic values and present them as historical client results.
- If portfolio dashboard screenshots are recreated, clearly label them as sanitized/reconstructed visuals when they are not original client screenshots.
- Public-comment observations used as qualitative context must remain separate from calculated internal-data findings.

13. Known Limitations / Open Design Decisions

Open Detail	Treatment
Exact historical dashboard page names/count	Not retained; use logical portfolio views only.
Exact DAX formulas	Document separately only when reconstructed from verified KPI logic; do not claim they are original historical measures.
Exact SQL implementation	Not retained; do not invent table/query details.
Exact inactivity threshold for churn	Not retained; requires confirmation before exact measure implementation.
Exact original visual theme/colors	Not retained; portfolio styling may be designed separately and labeled accordingly.
Refresh/security architecture	Not established from current evidence; do not claim production gateway, RLS or scheduled-refresh implementation.

14. Traceability

Power BI Requirement Area	Business / User Story Traceability
Retention & Attrition	BR-02, BR-03, BR-11 US-01
Cohort Analysis	BR-04, BR-11 US-02
Utilization	BR-05, BR-06 US-03
Renewal & Plan	BR-07, BR-08, BR-12 US-04
Trainer Utilization	BR-09 US-05
Consolidated Dashboard	BR-01, BR-13 US-06
Data Quality / Confidentiality	BR-14, BR-15 US-07